

Machine Learning Pipeline

Formative 2: Multimodal Data Preprocessing Assignment

Member	Role
Akinloye Emmanuel	Image processing and face recognition
Kevin Nizeyimana	Customer data merge, product model
Henriette Biziyaremye	Audio features and voice verification
Dorian Sheja Bitega	Model training, evaluation, command-line app

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Links:

- Contribution Tracker - Task_sheet
- GitHub Repository - Click here
- Video - Youtube

Table of Contents

1. Introduction
2. System Design
3. Data and Privacy
   3.1 Tabular Data
   3.2 Facial-Image Data
   3.3 Audio Data
   3.4 Consent and Privacy

4. Product Recommendation
   4.1 Data Preprocessing and Merging
   4.2 Exploratory Analysis
   4.3 Feature Engineering and Leakage Prevention
   4.4 Model Development
   4.5 Product-Model Results

5. Image Processing and Facial Recognition
   5.1 Face Detection and Alignment
   5.2 Image Augmentation
   5.3 Facial-Feature Extraction
   5.4 Face-Classification Model
   5.5 Face-Model Results
   5.6 Unknown-Face Rejection
   5.7 Final Unauthorized-Face Test

6. Audio Processing and Voice Verification
   6.1 Audio Preprocessing
   6.2 Audio Augmentation
   6.3 Voice Features
   6.4 Voice Model
   6.5 Voice Results

7. Multimodal Integration
   7.1 Application Implementation
   7.2 Access-Control Rule
   7.3 End-to-End Tests

8. Combined Results
9. Limitations and Ethical Considerations
10. Team Contributions
11. Conclusion
12. References

1. Introduction

This project develops a multimodal system that combines facial recognition, voice verification and product recommendation. The system first verifies a user's face, generates a product recommendation, and then verifies that the provided voice belongs to the same identity recognized by the face model. The recommendation is displayed only when both authentication checks succeed.

The project has three main objectives:

- Predict a product category from customer and transaction data.
- Recognize authorized team members while rejecting an unknown face.
- Verify that a voice belongs to the person identified by the face model.

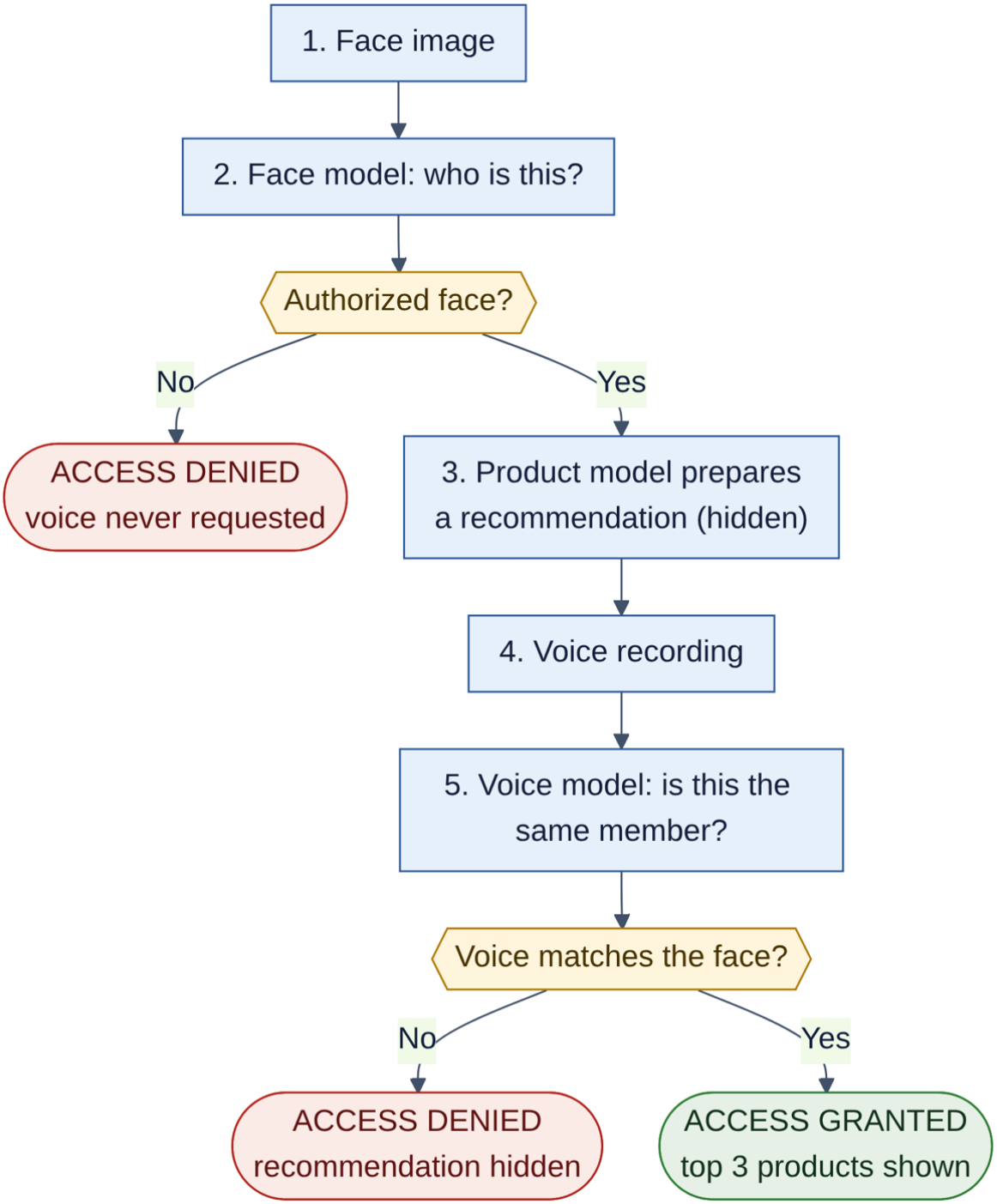
The system must reject an unauthorized face, an unauthorized voice, or a voice belonging to a different authorized member.

2. System Design

The intended system workflow is:

- Receive a facial image.
- Detect and align the face.
- Extract a facial embedding and predict the identity.
- Reject the image when facial similarity is below the fixed threshold.
- Generate a product recommendation but keep it hidden.
- Receive an audio recording.
- Verify that the voice matches the face-recognized identity.
- Show the recommendation only when both checks pass.

Figure 1: Final multimodal workflow



- The user gives a face image, and the face model decides whether it is an authorized face.
- If the face is not authorized, the result is ACCESS DENIED and the voice is never requested.
- If the face is authorized, the product model prepares the recommendation but keeps it hidden.
- The user then gives a voice recording, and the voice model checks whether it matches the same member as the face.
- If the voice does not match, the result is ACCESS DENIED and the recommendation stays hidden. If it matches, the result is ACCESS GRANTED and the top 3 products are shown.

3. Data and Privacy

3.1 Tabular data

The project uses two tabular files we were given, customer_social_profiles.csv has 155 rows and 5 columns (social customer ID, platform, engagement score, purchase-interest score, and review sentiment) and describes 84 unique customers. customer_transactions.csv has 150 rows and 6 columns (customer ID, transaction ID, amount, date, product category, and rating) and covers 75 unique customers, dated January to May 2024.

Both files repeat customers, so neither is one row per person. The two files also write the customer ID differently: the social file uses an A prefix such as A178, while the transaction file uses 178. After normalising the ID, 61 customers appear in both files.

The prediction target is product_category, which has five balanced classes: Sports (28 rows), Electronics (27), Clothing (22), Books (20), and Groceries (20). The final merged dataset contains 117 rows and 18 columns, which covers about 61 customers at an average of 1.92 purchases per customer.

3.2 Facial-image data

Four authorized members each supplied three original facial images representing neutral, smiling and surprised expressions. This produced 12 authorized original images.

A separate unauthorized identity supplied three images for threshold validation. A different unauthorized identity supplied three images for the final held-out test. The final unauthorized-test images were not used for model training or threshold selection.

3.3 Audio data

Four authorized members each recorded two phrases, "Yes, approve" and "Confirm transaction", three times each. This produced 24 authorized original recordings (six per member, three takes of each phrase). Four people outside the team each recorded both phrases once, giving 8 unauthorized original recordings from four different unauthorized speakers.

The recordings were made on members' own phones and arrived in three formats: MP4 (video), AAC and OGG. All of them were converted to 16 kHz mono WAV with src/convert_audio.py before any

processing, and the video track of the MP4 files was discarded. Every file follows one naming rule, <member_id>_<phrase>_<take> (for example member01_yes_approve_2), using the same member IDs as the image pipeline so that the voice check can be compared with the identity recognised by the face model.

3.4 Consent and privacy

Facial images and voice recordings are biometric data. Participants provided
permission for their data to be used in this student project. Raw facial images were kept outside the public repository. The repository contains the consented audio recordings required for voice-model reproducibility. Access to biometric data should remain restricted, and all participants must have consented to its repository storage and use.

4. Product Recommendation

Section owner: Kevin Nizeyimana

4.1 Data preprocessing and merging

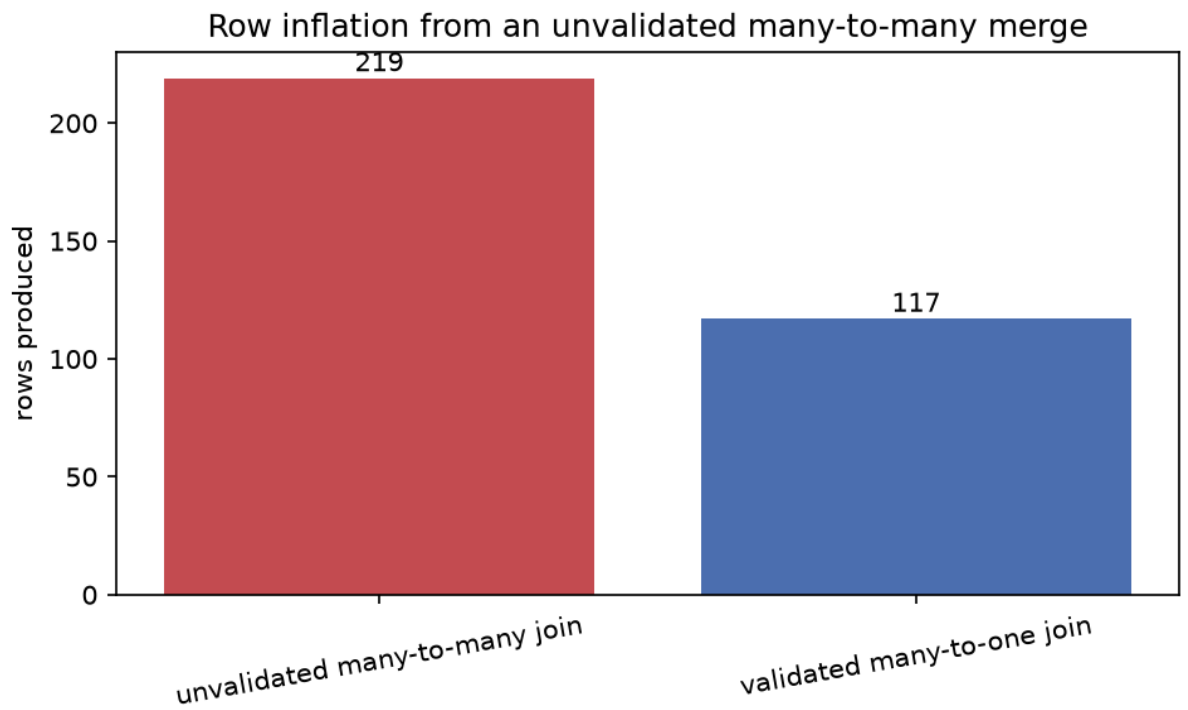
ID normalisation. Both identifiers were converted to text, trimmed, lowercased and stripped of the leading A, then stored as one shared integer customer_id. This created a single key that can be used for validation and for grouping the train/test split.

Duplicates and missing values. The 155 social rows reduce to 84 customers, and the 150 transaction rows to 75 customers, because customers buy more than once. The social table therefore had to be summarised to one profile per customer before joining; otherwise the join multiplies rows. After merging, no two rows are exact duplicates.

The merged table contains 254 blank cells, and all of them are expected rather than errors. Every customer's first purchase has no earlier purchase to describe, so prior_avg_amount, prior_mode_category and days_since_prev are empty on those 61 rows; prior_avg_rating is empty on 63 rows and customer_rating is missing on 8 purchases. No rows were deleted for this reason; the model’s imputer fills the gaps during training.

Merge validation. The relationship was declared with validate "many_to_one" so a failure would raise an error instead of silently duplicating rows. A raw join without that check produces 219 rows instead of 117, a 1.87× inflation; the validated merge gives exactly 117 rows for 117 real sales. Of the 84 social customers, 23 have no transaction and were dropped, and 33 transactions belong to customers with no social profile and were also dropped, because the product model needs both halves.

Result. data/processed/merged_dataset.csv 117 rows × 18 columns, 61 customers.



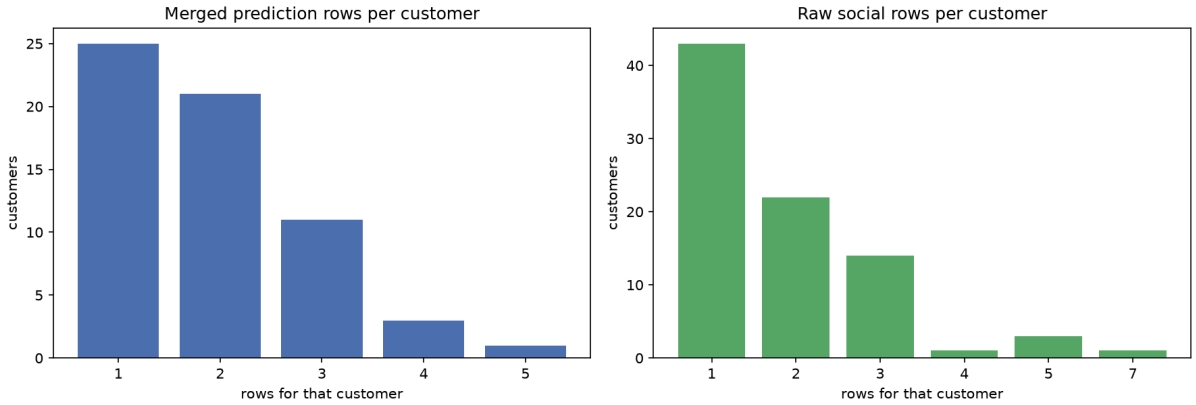
Row inflation from an unvalidated join. The raw many-to-many merge produces 219 rows from 117 real transactions

4.2 Exploratory analysis

Three findings shaped the modelling decisions.

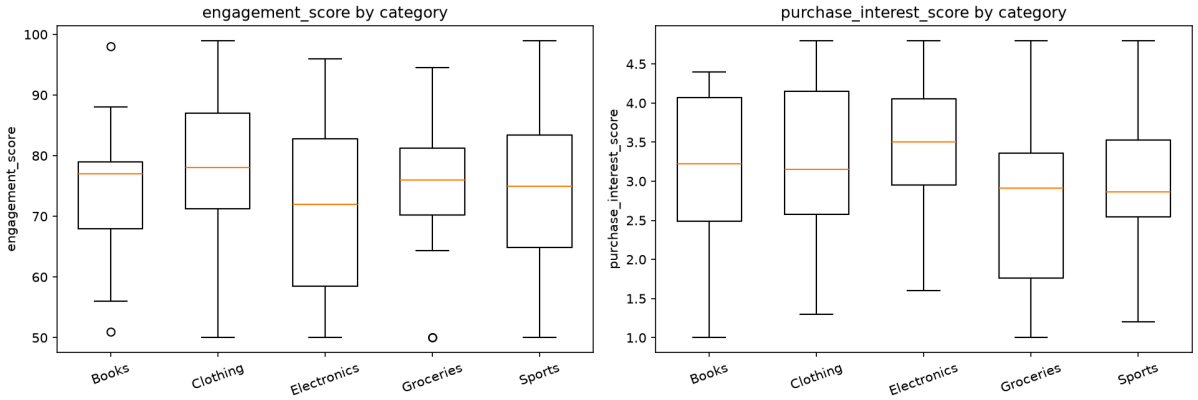
1. The target is balanced but small. Sports 28, Electronics 27, Clothing 22, Books 20 and Groceries 20. With five classes, a model that always guesses the most common class scores about 20%. That number is the real baseline, and any accuracy near it means the model has learned nothing.

2. Customers buy different numbers of times. 25 customers appear once, 21 twice, 11 three times, 3 four times and 1 five times. Because the same person contributes several rows, a random train/test split would place the same customer on both sides and let the model memorise them. This is why the split is grouped by customer.



Rows per customer in the raw social file (right) and in the merged prediction table (left).

3. The social signals barely move with the target. Average engagement and purchase interest are almost identical across the five categories (engagement 71.3 to 77.3, interest 2.69 to 3.49), and no numeric column correlates with purchase amount above |r| = 0.19. Purchase amount itself has a mean of 287.91, a median of 280, a range of 62 to 495 and no values beyond two standard deviations, so there are no extreme outliers to explain. In short, the data is clean but the relationship between the social profile and the product category is weak which the results later confirm.



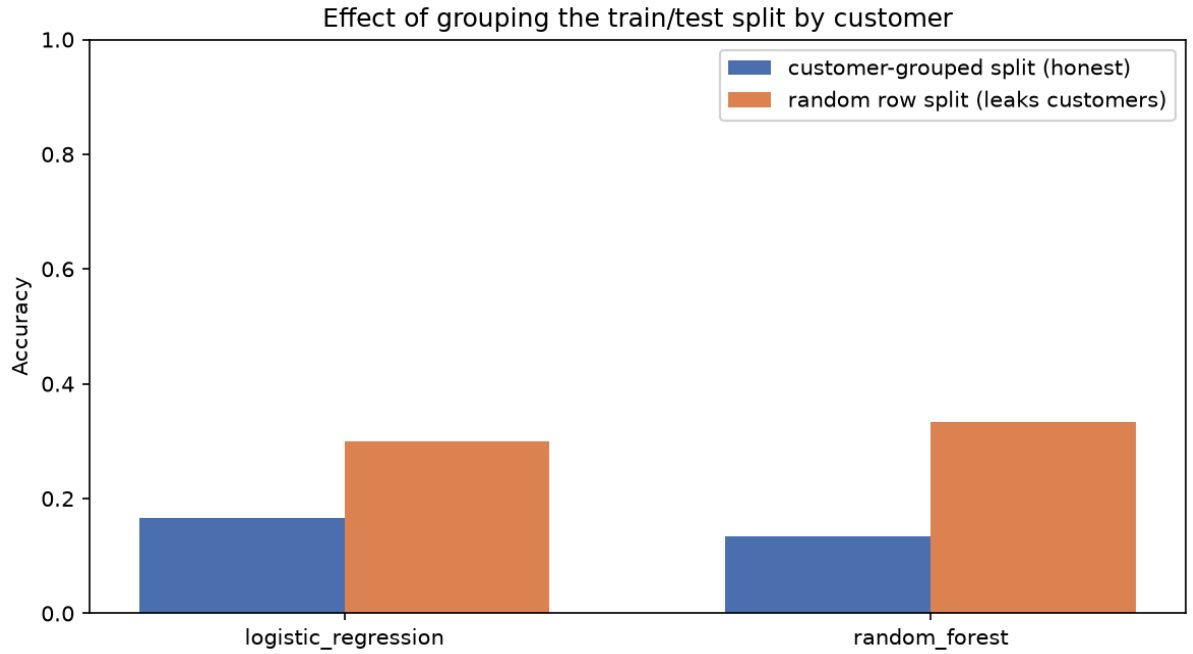
Engagement score and purchase-interest score by product category. The distributions overlap, so the social signals do not separate the classes.

4.3 Feature engineering and leakage prevention

Twelve features were used: nine numeric (engagement score, purchase interest, number of social records, number of earlier transactions, average earlier amount, average earlier rating, days since the previous purchase, month, day of week) and three categorical (platform, review sentiment, most frequent earlier category).

Leakage prevention. The spending, rating and category history columns are computed only from purchases made before the row being predicted. purchase_amount and customer_rating are recorded after the sale, so they are excluded from the features entirely. Including them would let the model read part of the answer. transaction_id and customer_id are excluded as identifiers, and purchase_date is replaced by month, day of week and days since previous purchase.

Grouped splitting. The data was split 75/25 by customers using GroupShuffleSplit from scikit-learn [1], giving 87 training rows from 45 customers and 30 test rows from 16 customers, with zero customers in both sets. A random split of the same data puts 19 of the 61 customers on both sides, which is why its scores are reported only as a warning, not as the result.



Grouped split versus random split. The random split places 19 of 61 customers in both sets and inflates every score.

4.4 Model development

All models were implemented using scikit-learn pipelines [1]. A ColumnTransformer fills missing numbers with the median and scales them, fills missing categories with the most frequent value and one-hot encodes them (ignoring unseen categories), and then passes the result to the classifier. Keeping preprocessing inside the pipeline means the saved file can be applied to a new customer without repeating any manual steps.

Four models were compared: a majority-class baseline, a rule baseline that simply reuses the customer's most frequent earlier category, Logistic Regression (C = 1.0, 5000 iterations) and a Random Forest (300 trees, minimum 2 samples per leaf, fixed random state 42). Logistic Regression was selected because it had the best macro F1-score on the grouped hold-out and is the simpler model; it was saved as models/product_model.joblib.

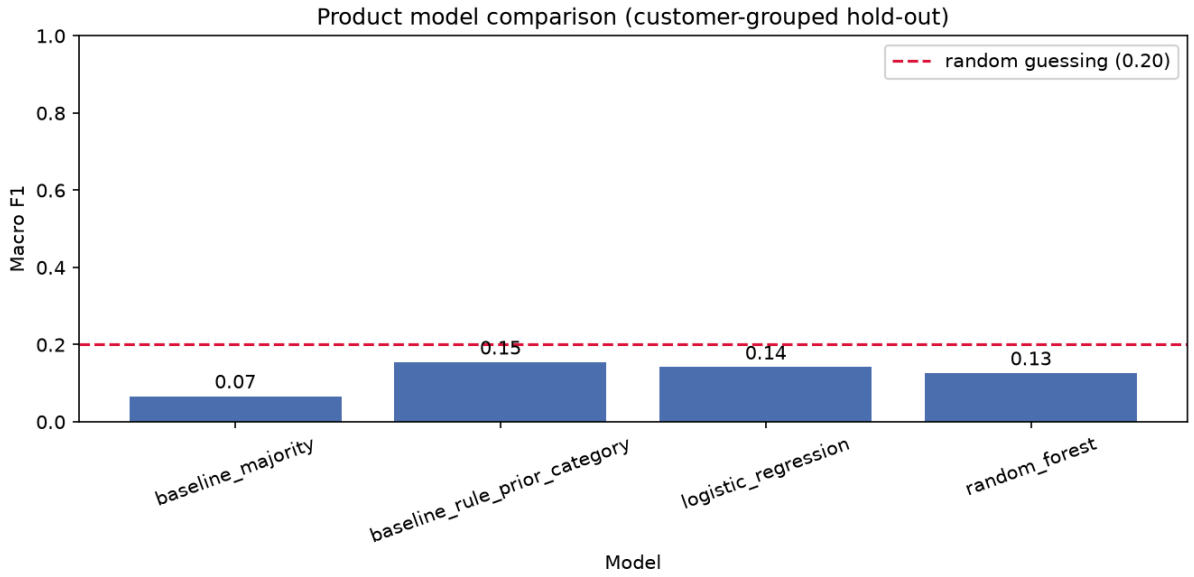
A small helper, predict_product(row, top_k=3), loads the saved pipeline and returns the three most likely categories with their probabilities, so the integration layer can show a ranked recommendation.

4.5 Product-model results

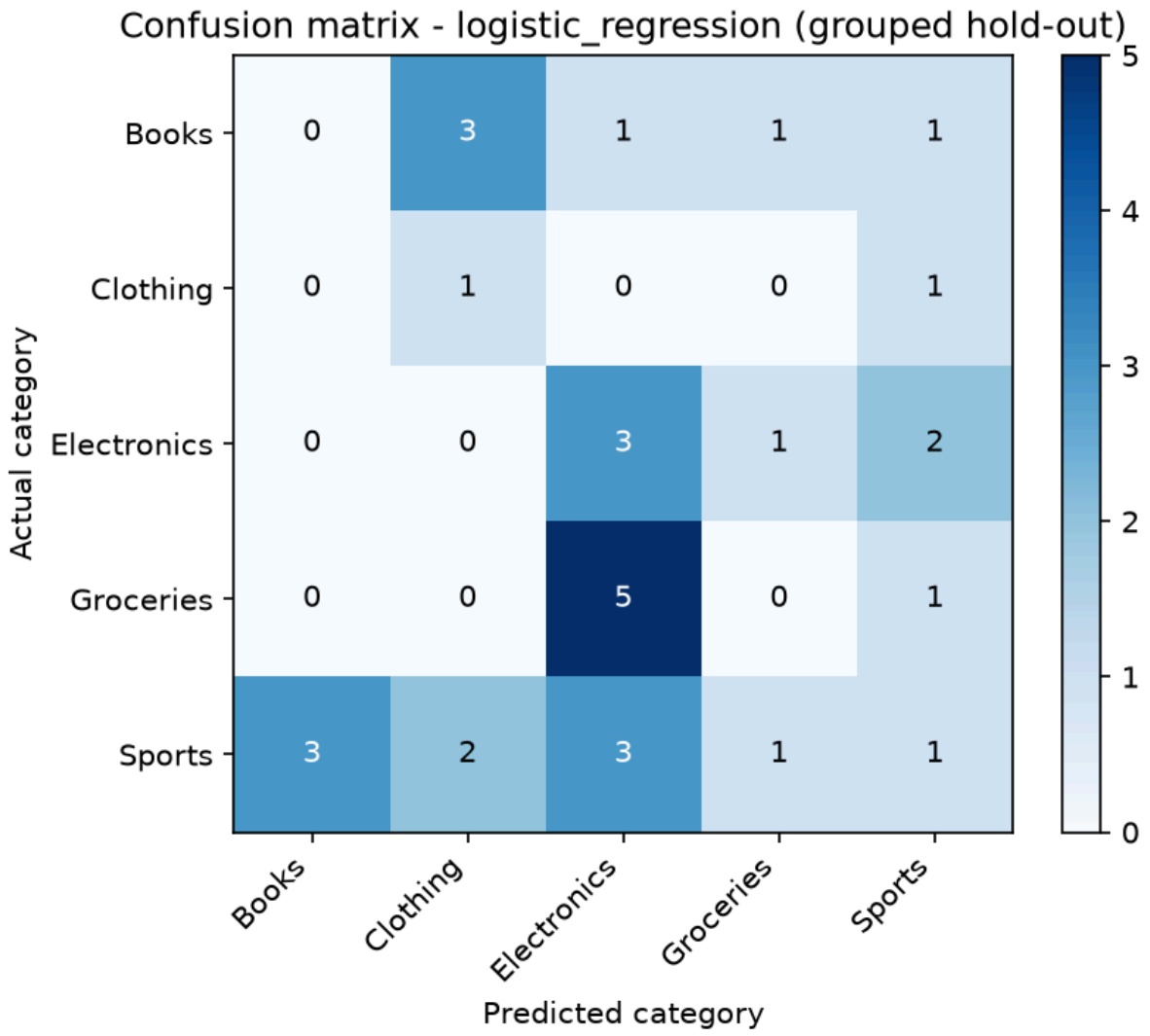
Metric	Result
Accuracy	0.167
Macro F1-score	0.142
Weighted F1-score	0.125
Log loss	2.751

These are the scores of the saved Logistic Regression model on the customer-grouped hold-out set of 30 rows. For comparison: the majority baseline scores 0.200 accuracy with a macro F1 of 0.067, the rule baseline 0.200 with 0.155, and the Random Forest 0.133 with 0.127 and a lower log loss of 1.813. The selected model is therefore below the majority baseline on accuracy.

This is the honest result and it is reported as a limitation, not as a success. With 87 training rows, five categories and near-identical social signals across those categories, there is no learnable pattern, so the model performs at chance level (0.200). A random split would report 0.300 for Logistic Regression and 0.333 for Random Forest, but 19 of 61 customers would appear in both sets, so those numbers measure memorisation rather than prediction and are shown only to demonstrate the effect of leakage. The confusion matrix supports the same conclusion: the model never predicts Books or Groceries correctly.



Macro F1-score by model on the customer-grouped hold-out set.



Confusion matrix of the saved product model on the 30 held-out rows.

5. Image Processing and Facial Recognition

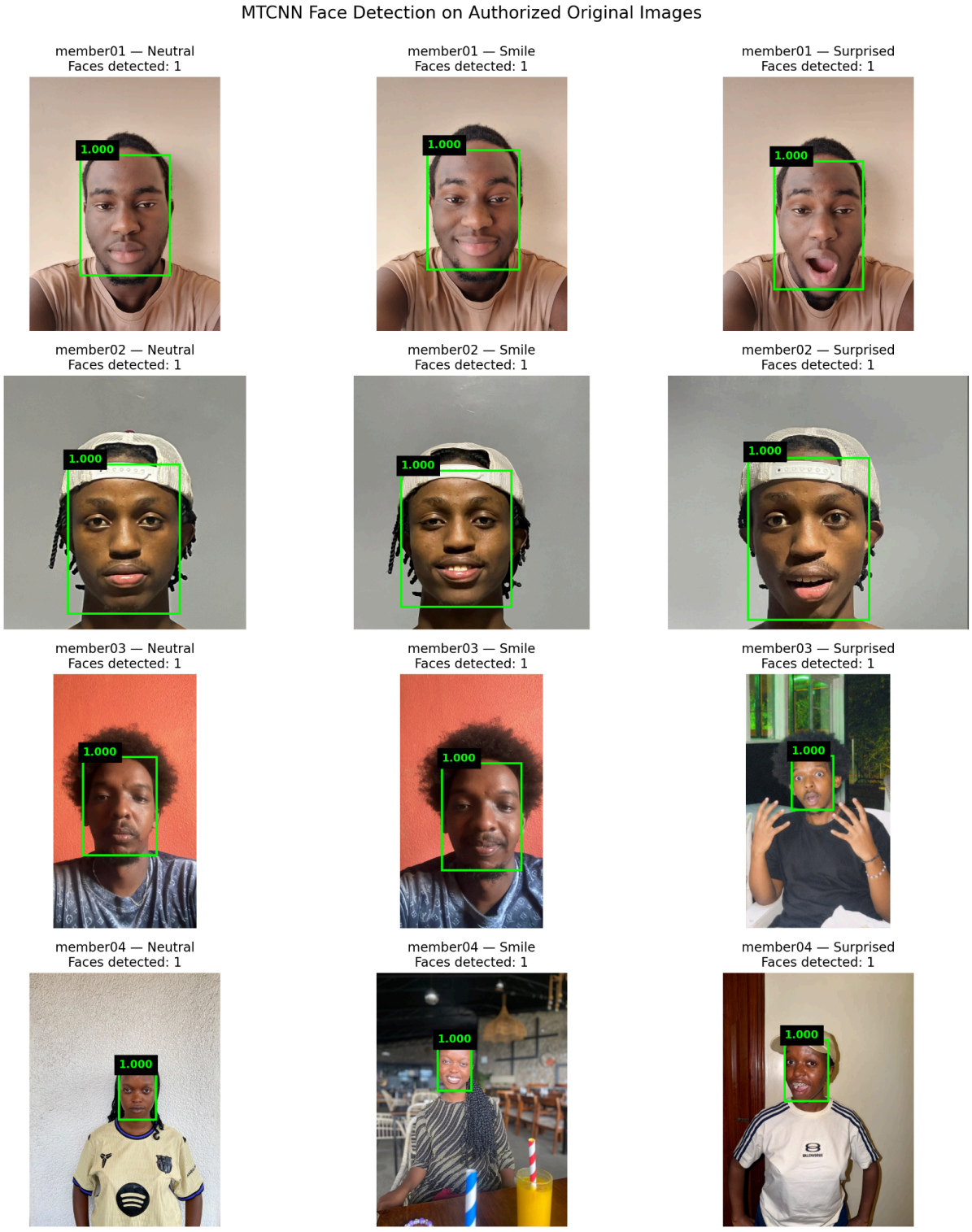
Section owner: Akinloye Emmanuel

5.1 Face detection and alignment

MTCNN was used to detect, align and crop each authorized face [2]. Images were corrected for EXIF orientation, converted to RGB and resized to 160 × 160 pixels, which is the input size expected by the FaceNet embedding model.

Exactly one face was detected in every authorized original image. Detection confidence exceeded 0.999 for all 12 images.

Figure 2: MTCNN detections on authorized images



5.2 Image augmentation

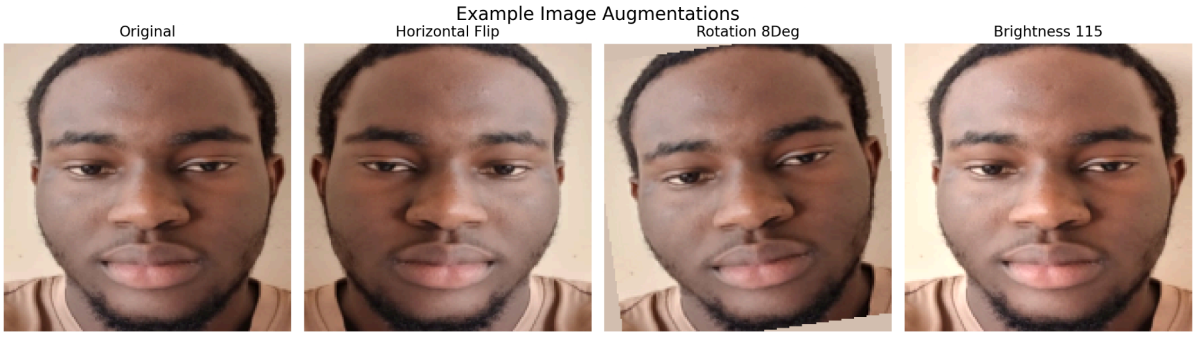
Each aligned image produced four processed variants:

- Original aligned image

- Horizontal flip
- Eight-degree rotation
- Brightness increased to 115%

The 12 originals therefore produced 48 processed images. Augmentation was used only for training. Augmented versions of test images were excluded from model evaluation.

Figure 3: Examples of image augmentation



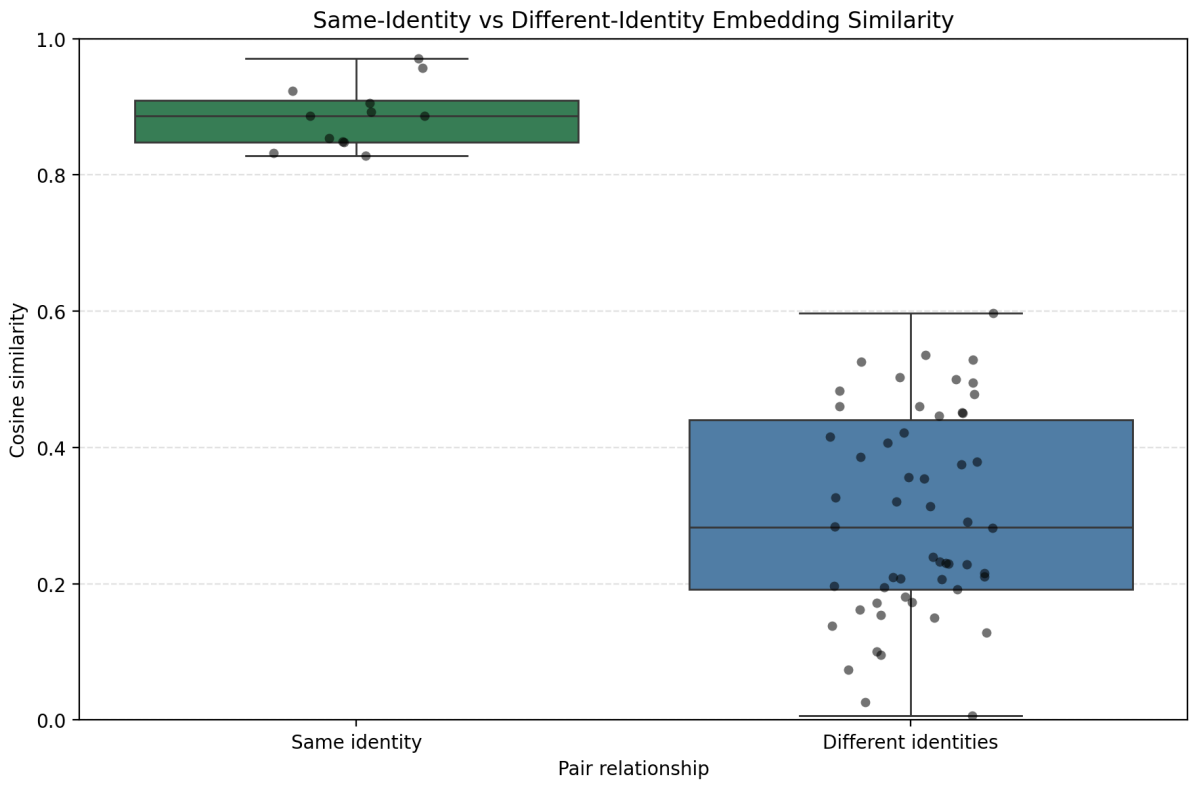
5.3 Facial-feature extraction

A FaceNet InceptionResnetV1 model [3], pretrained on the VGGFace2 dataset [4], converted each processed face into a normalized 512-dimensional embedding. The final feature table contained 48 rows, six metadata columns and 512 embedding columns. It contained no missing embedding values or duplicate sample identifiers.

Cosine similarity analysis showed clear separation between identities:

Relationship	Mean similarity	Minimum	Maximum
Same identity	0.886153	0.828432	0.970645
Different identities	0.299747	0.006728	0.597566

Figure 4: Same-identity and different-identity similarity



5.4 Face-classification model

A pipeline containing `StandardScaler` and Logistic Regression was trained on the facial embeddings.

Evaluation used three-fold leave-one-expression-out validation. Each fold held out one expression, neutral, smile or surprised, across all four members.
The remaining two expressions and their augmentations were used for training.
Testing used only the four held-out original images.
This design ensured that an original test image and its augmented variants could not appear in training.

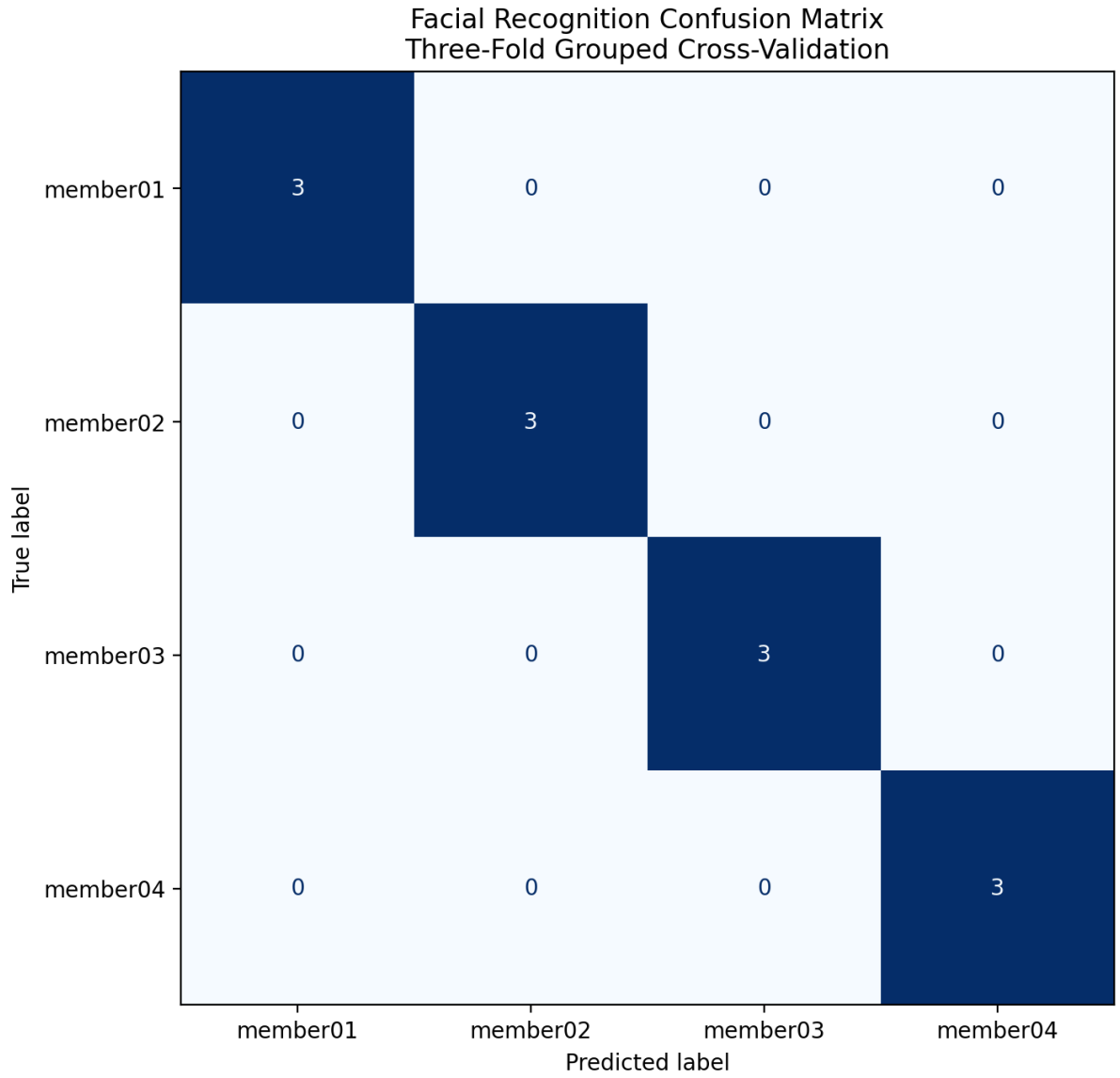
5.5 Face-model results

Metric	Result
Authorized Originals evaluated	12
Accuracy	1.000
Macro F1-score	1.000
Weighted F1-score	1.000

Log loss	0.005542

All 12 held-out original images were classified correctly.

Figure 5: Facial-recognition confusion matrix



These results demonstrate strong separation within the collected dataset. However, the dataset is too small and controlled for the scores to be interpreted as real-world biometric-security performance.

5.6 Unknown-face rejection

A closed-set classifier always assigns an input to one of its known identities. Therefore, classifier probability was not treated as authentication confidence.

An additional cosine-similarity check was applied between the input embedding and the predicted identity centroid.

The threshold was selected using authorized leave-one-out scores and the separate unauthorized-validation identity:

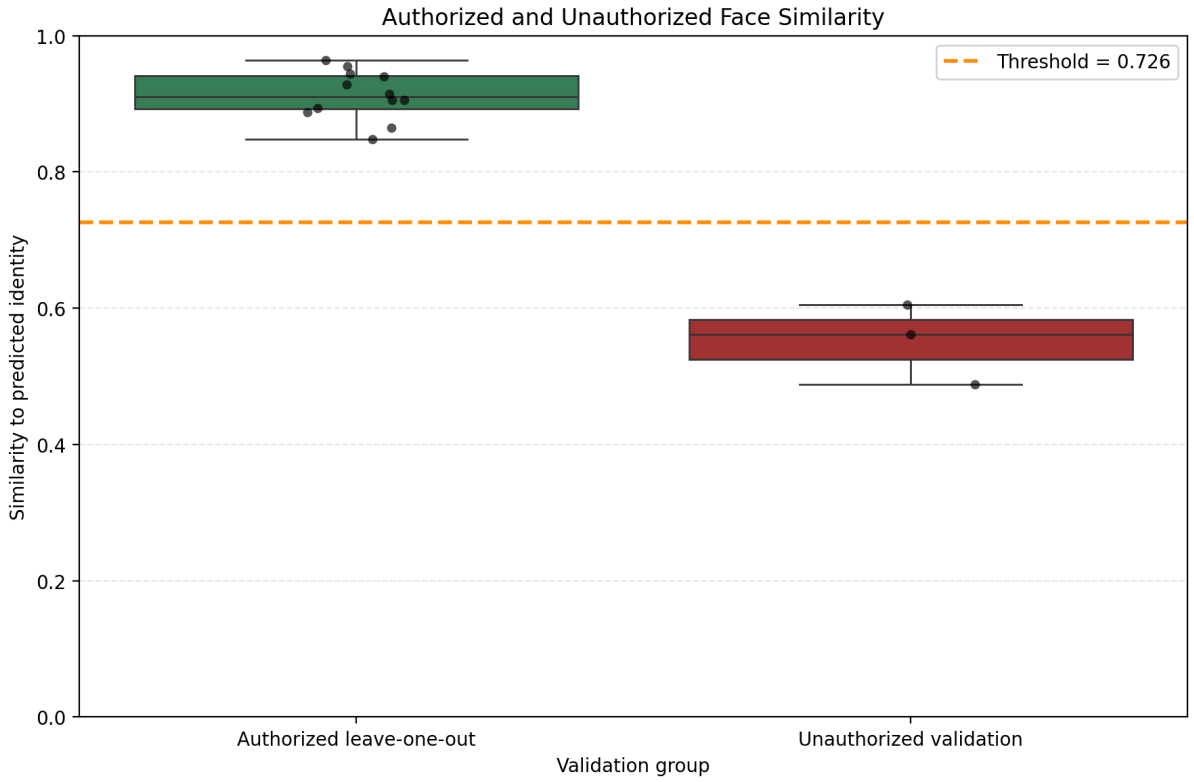
Threshold measurement	Value
Minimum authorized similarity	0.847625
Maximum unauthorized-validation similarity	0.605112
Separation gap	0.242512
Selected threshold	0.726368

The threshold was the midpoint between the minimum authorized similarity and maximum unauthorized-validation similarity.

At this threshold:

Validation result	Value
Authorized false rejection rate	0.000
Unauthorized-validation false acceptance rate	0.000

Figure 6: Authorized and unauthorized similarity with threshold



5.7 Final unauthorized-face test

The fixed threshold was evaluated on a different unauthorized identity whose images had not been used for training or threshold selection.

Expression	Similarity	Decision
Neutral	0.512213	Rejected
Smile	0.481500	Rejected
Surprised	0.390853	Rejected

All three images were rejected, producing a final test false acceptance rate of 0.000.

The closed-set classifier assigned these images to Member 4 and produced high classifier confidence for two of them. The similarity threshold nevertheless rejected them. This demonstrates why classifier probability alone is not a suitable authentication criterion.

The selected threshold is a validation-derived experimental threshold, not a production-security threshold. Only one unauthorized identity was available for

validation and one different identity was available for final testing.

6. Audio Processing and Voice Verification

Section owner: Henriette Biziyaremye

6.1 Audio preprocessing

Every recording was processed using the same pipeline in `src/preprocess_audio.py`, ensuring that new voice samples were handled consistently with the training data. Audio loading, resampling, silence trimming and feature processing were implemented using librosa [5]. The preprocessing pipeline consisted of the following four steps:

- Resampling to 16 kHz. Phone recordings arrived at different sample rates. Speech information lies mostly below 8 kHz, so 16 kHz keeps it while making every file comparable.
- Mono conversion. Stereo files were averaged into one channel, because the second channel adds no speaker information.
- Silence trimming. Leading and trailing silence more than 30 dB below the loudest point was removed, so features describe the voice rather than the pause before someone starts speaking.
- Peak normalisation. Each clip was scaled so its loudest sample equals 1. This stops differences in phone volume, or how close someone held the phone, from dominating the features.

For each member, both phrases were plotted as a waveform and a mel spectrogram in notebooks/03_audio_processing.ipynb. The plots and the measurements taken from them show clear differences between speakers. Member03 has a clearly higher voice (about 196 Hz median pitch), while the other three sit close together between about 83 Hz (member04) and 111 Hz (member01). Because three members share a similar pitch range, pitch alone cannot separate them, which is why the model relies on MFCCs, which describe the whole shape of the voice spectrum.

Speaking rate also varies: member04 is the fastest (0.8 s for "Yes, approve") and member01 the slowest (2.1 s for "Confirm transaction", with 45% of frames silent). In the spectrograms most energy sits below about 1 kHz. The "Yes, approve" clips of member02 and member03 have a much higher spectral centroid (about 2.7 kHz versus 1.2 to 1.5 kHz for the others) because both speakers draw out the "s" in "yes", which appears as bright energy between 2 and 8 kHz. The "p" in "approve" shows as a thin vertical streak across all frequencies. The pauses between words are dark, so the recordings themselves are clean.

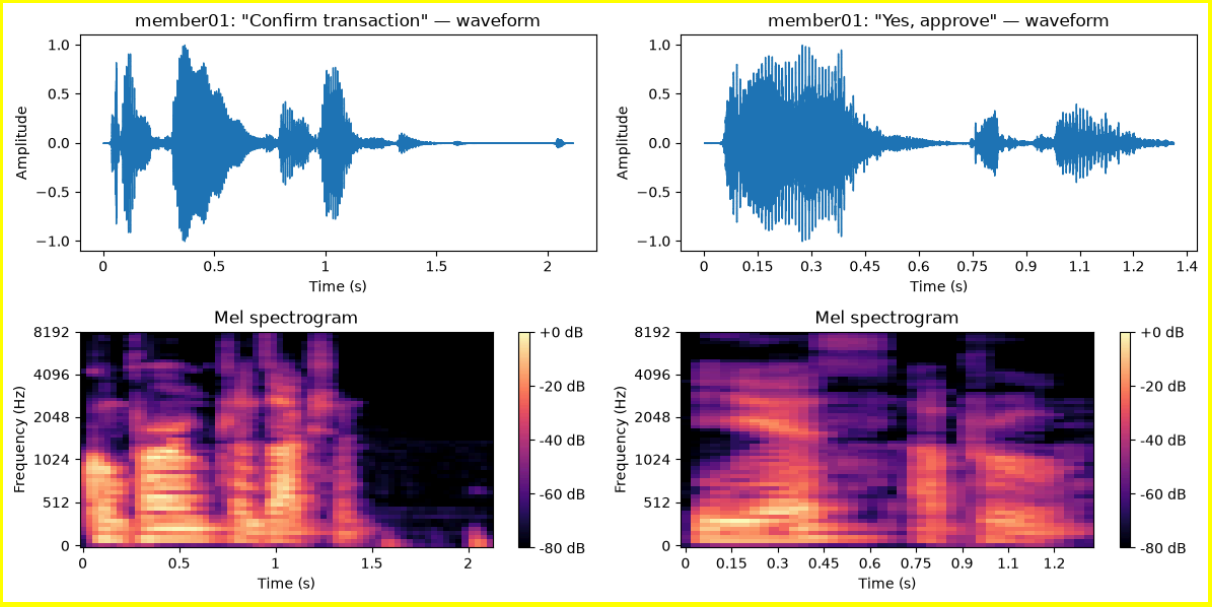


Figure 7: Waveforms and mel spectrograms for member01's "Confirm transaction" and "Yes, approve" recordings.

6.2 Audio augmentation

Each original recording was augmented three times so that the voice model sees realistic variation and does not memorise a single recording:

Augmentation	Setting	What it simulates
Pitch shift	+2 semitones	A slightly higher voice (tiredness, a cold, excitement)
Time stretch	1.15× speed, pitch unchanged	Speaking faster
Background noise	White noise at 20 dB SNR	A noisy room or a low-quality microphone

Each original therefore yields four versions (the original plus three augmentations): 96 member rows from 24 recordings and 32 unauthorized rows from 8 recordings. An original and its augmented copies always stay in the same data split, so an augmented copy of a test recording can never appear in training.

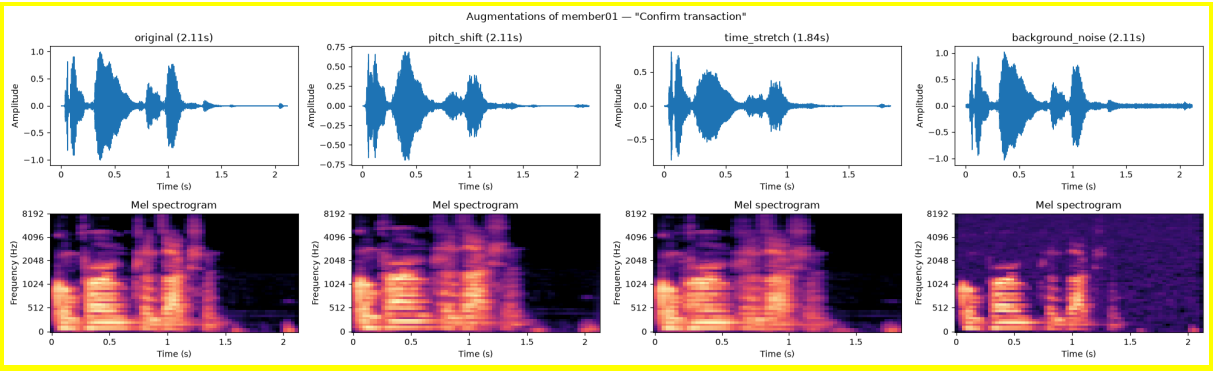


Figure 8: Waveform and mel-spectrogram comparison of the original recording and its pitch-shifted, time-stretched and background-noise augmentations.

6.3 Voice features

Each clip is turned into one fixed-length row of 35 numbers by computing features over short frames and summarising them with their mean and standard deviation. The features are saved in data/processed/audio_features.csv (members) and data/processed/audio_features_unauthorized.csv (outsiders).

Feature	Columns	What it captures
MFCC 1–13 (mean and standard deviation)	26	The shape of the vocal tract; the core of the voiceprint
Spectral roll-off (mean, std)	2	Frequency below which 85% of the energy lies
Spectral centroid (mean, std) and bandwidth (mean)	3	Brightness and spread of the voice
RMS energy (mean, std)	2	Loudness and how it changes
Zero-crossing rate (mean)	1	Noisiness; voiced versus unvoiced sounds
Duration	1	Speaking rate

The final feature tables contain no missing values. The same extract_features() function is imported by the application, so a live voice sample is described exactly like the training data.

6.4 Voice model

The selected method used a pretrained ECAPA-TDNN speaker embedding [6], implemented through SpeechBrain [7] and trained on the VoxCeleb speaker dataset [8]. Each recording was scored using cosine similarity to the claimed member's enrollment centroid.

Data splits. The splits were made so that no recording, and no outsider, appears in more than one set:

Split	Members (genuine)	Outsiders (unknown)
Training	Take 1 of each phrase (8 originals, 32 rows)	None; the model never trains on outsiders
Validation	Take 2 of each phrase (8 originals, 32 rows)	2 speakers (4 originals, 16 rows)
Test	Take 3 of each phrase (8 originals, 32 rows)	2 different speakers (4 originals, 16 rows)

Verification against the face identity. The voice step is treated as verification, not open identification. The face model first names a member (the claimed identity), and the voice is accepted only if its score for that member reaches the selected threshold. A voice from another member, or from an outsider, is rejected even if it resembles someone in the team. The centroid is built from each member's training recordings (take 1 plus its augmentations). The voice is accepted only when its similarity to the claimed member is at least the threshold and that member is also the closest match among the four. Three methods were compared on validation only:

- logistic regression with confidence and distance checks
- cosine similarity on hand-crafted MFCC/spectral features
- cosine similarity on ECAPA embeddings

ECAPA was chosen because its validation FRR was 6.2%, against 53–88% for the other two, with FAR at or below 5%.

Threshold selection. The threshold was chosen using only the validation members and validation outsiders, with this rule: the lowest false rejection rate (FRR) with the false acceptance rate (FAR) for outsiders at or below 5% and the member-mismatch rate at or below 5%. When several thresholds tie, the middle of that range is used. If no threshold met both limits, the equal-error-rate threshold would be used instead. The selected cosine-similarity threshold is 0.345. The test members and the two test outsiders were used once, after the threshold was fixed.

6.5 Voice results

Three error rates describe the verification step. The false acceptance rate (FAR) is the share of outsider attempts that were accepted. The false rejection rate (FRR) is the share of members rejected when claiming their own identity. The identity-mismatch rate is the share of members accepted when claiming a different member’s identity.

Metric	Validation	Test
Accuracy (member identification)	1.000 (100%)	1.000 (100%)

Macro F1-score	1.000	1.000
Weighted F1-score	1.000	1.000
Log loss	0.076	0.133
Selected voice threshold	0.345 (cosine similarity, ECAPA)	Same (fixed on validation)
False acceptance rate (outsiders)	0.0% (0/16)	6.2% (1/16)
False rejection rate (members)	0.0% (0/8)	0.0% (0/8)
Identity-mismatch rate	0.0% (0/24)	0.0% (0/24)

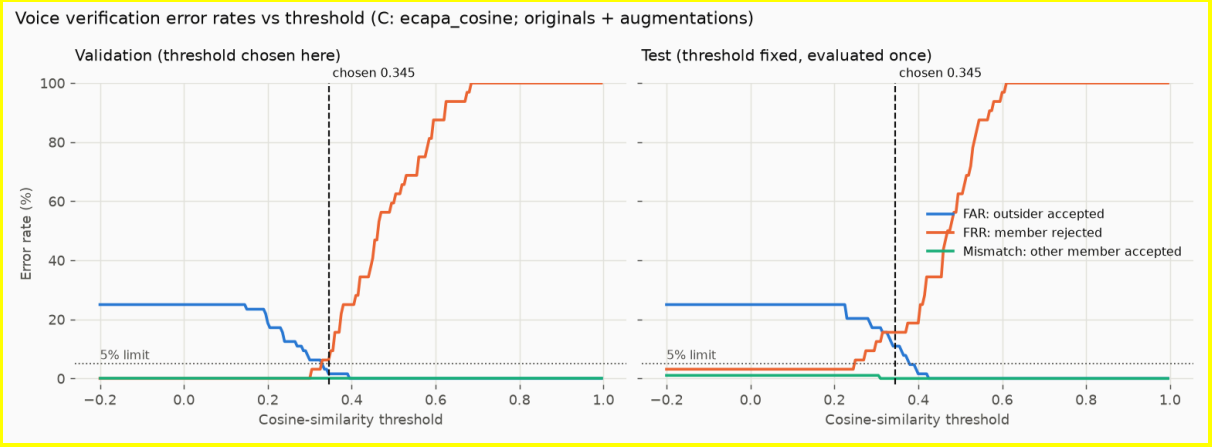


Figure 9: Voice-verification error rates across cosine-similarity thresholds for the validation and held-out test sets. The threshold of 0.345 was selected using validation data and then fixed for final testing.

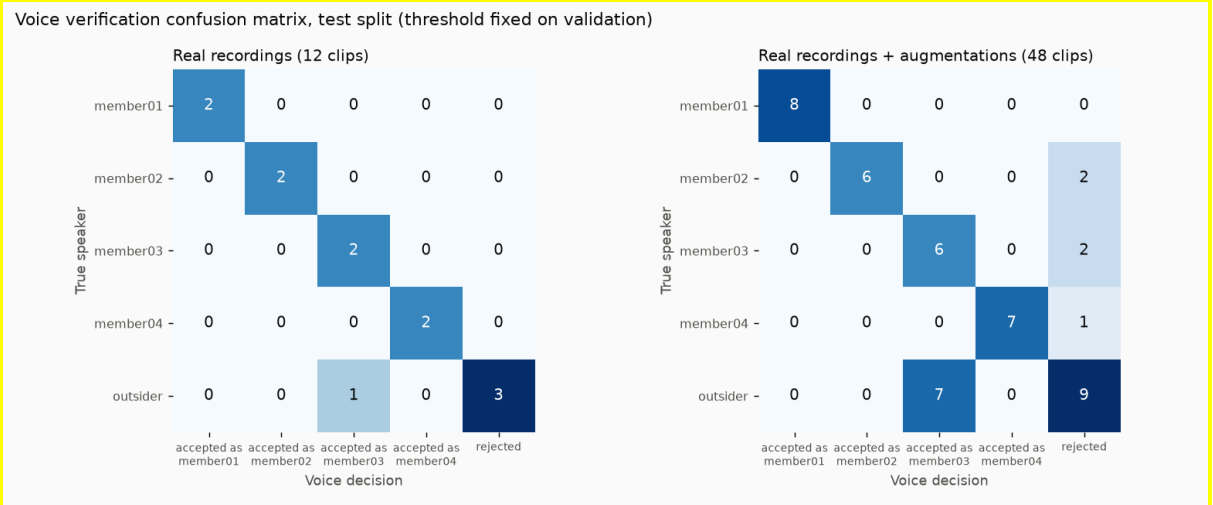


Figure 10: Confusion matrix for member identification on the held-out original voice recordings

On the real test recordings, 1 of the 16 outsider attempts was accepted and none of the 8 genuine attempts was rejected. member03 was the weak point. The single accepted outsider attempt was incorrectly accepted as member03, indicating that member03's enrollment centroid was the weakest

point in this small test, so their voice profile seems to be the closest to the outsiders. That is a 6.2% false acceptance rate, down from 59.4% with the earlier model.

These results come from a small dataset: three takes per phrase for each member and four outsiders, only two of whom were used for testing. One more accepted outsider attempt would raise the test FAR by about six percentage points. The numbers show the method works on this data, but they don't measure real-world security.

7. Multimodal Integration

Section owner: Dorian Sheja Bitega

7.1 Application implementation

The file src/app.py is the command-line program that puts our three models together. It uses three saved models from the models/ folder: the **face model**, the **voice model** and the **product recommendation model**.

The program works in four steps:

1. **Face check.** The user gives a photo. The face model says which member it is, or says it is unknown.
2. **Recommendation prepared.** If the face is accepted, the product model prepares a recommendation, but the program does not show it yet.
3. **Voice check.** The user gives a voice recording. The voice model checks that the voice belongs to the same member that the face model found.
4. **Result.** If both checks pass, the program shows the top three recommended products. If not, it prints ACCESS DENIED and the reason.

If a file is missing or has the wrong format, the program prints a clear message and denies access. It does not crash. To run it: python src/app.py --face <photo> --voice <recording>.

7.2 Access-control rule

The recommendation is shown only when **both** conditions are true:

- the face is recognised as an authorized member, and
- the voice belongs to **that same member**.

If the face is not recognised, access is denied at once and the voice is never requested. If the face is correct but the voice belongs to a different member, access is also denied. In every denied case, the recommendation is never printed.

7.3 End-to-end tests

Scenario	Expected result	Actual result
Authorized face and matching voice	Recommendation displayed	**Granted.** Face and voice were verified as member01. Recommendations: Books (31%), Groceries (24%) and Sports (17%).
Unauthorized face	Access denied before voice verification	**Denied.** The face was not recognized, so the voice-verification stage was not reached.
Authorized face and unauthorized voice	Recommendation remains hidden	**Denied.** The face was accepted, but the outsider's voice did not match member01.
Authorized face and another member's voice	Identity mismatch; recommendation remains hidden	**Denied.** The face matched member01, but the voice matched member02.

Example of a successful attempt

Face recognised as member01 (confidence 1.00).

A recommendation is ready. Voice verification is required to display it.

Voice verified as member01 (confidence 0.54).

ACCESS GRANTED. Recommended products:

1. Books (31%)
2. Groceries (24%)
3. Sports (17%)

Example of a rejected attempt (member01's face with member02's voice)

Face recognised as member01 (confidence 1.00).

A recommendation is ready. Voice verification is required to display it.

ACCESS DENIED: voice sounds like member02, not member01 (confidence 0.42).

How the tests were run.

The four scenarios were first validated programmatically in notebooks/04_model_evaluation.ipynb using the saved models and decision rules. They were then executed through src/app.py using local

face images and held-out voice recordings. The raw demonstration face images were kept outside version control.

Limitations

- The voice model is not perfect. One of 16 outsider claimed-identity verification attempts was accepted. The test used only 2 outsider speakers, so this rate is uncertain.
- The product model is weak (accuracy 0.17, while random guessing gives 0.20), so the recommendations are only examples.
- Each member is linked to one customer from the merged dataset for the demo, because the project has no table that links members to customers.

8. Combined Results

Component	Accuracy	Macro F1-score	Log loss	FAR	FRR
Product recommendation	0.167	0.142	2.751	N/A	N/A
Face recognition	1.000	1.000	0.005542	0.000 (0/3)	0.000 (0/12)
Voice verification	1.000	1.000	0.133	0.063 (1/16)	0.000 (0/8)

The product model sits at chance level (0.200) because the tabular data contains no strong relationship between social behaviour and product category; the face and voice models reach high scores only on very small, controlled samples. Neither result should be read as real-world performance.

The results must be interpreted in relation to the small datasets. High controlled-dataset scores do not guarantee reliable performance under different cameras, microphones, lighting, poses, accents or environmental noise.

9. Limitations and Ethical Considerations

The main limitations are:

- Only four authorized identities were available.
- Each member supplied only three original facial expressions.
- Face validation and final testing each used one unauthorized identity.
- The voice dataset contains a small number of original recordings.
- Product-model performance is limited by the small tabular dataset and weak

predictive relationship between the available features and target.
- The system was evaluated under controlled conditions rather than realistic deployment conditions.
- Experimental authentication thresholds must not be treated as production security guarantees.

Facial and voice data can expose sensitive identity information. A real system would require informed consent, access controls, encryption, retention limits, secure deletion, bias testing and compliance with applicable data-protection requirements.

10. Team Contributions:

We have added the detailed contribution to the task sheet.

Team Member	Contribution
Kevin Nizeyimana	Tabular-data audit, EDA, merge validation, feature engineering and product-recommendation model.
Akinloye Emmanuel	Image preprocessing and augmentation, facial-feature extraction, face-classification and unknown-face rejection, repository coordination, CLI compatibility testing and final demonstration video.
Henriette Biziyaremye	Audio preprocessing, augmentation, audio-feature extraction and voice-verification model
Dorian Sheja Bitega	Combined evaluation, multimodal integration, CLI application, final testing.

11. Conclusion

This project demonstrates how product prediction can be combined with facial and voice authentication. The face pipeline successfully classified all 12 held-out authorized images and rejected all three images from the final unauthorized identity using a validation-derived similarity threshold.

The voice model, rebuilt around a pretrained speaker embedding with a threshold tuned only on validation data, accepted every genuine test recording, never confused one member for another, and accepted 1 of 16 outsider attempts, compared with 59.4% of unknown-voice attempts before. Its performance dropped on noisy and distorted clips, which shows that voice verification is the most sensitive part of the system to recording conditions.

The end-to-end application successfully integrated facial recognition, voice verification and product recommendation into a single authentication workflow. Testing confirmed that recommendations were displayed only when the face and voice belonged to the same authorized member. The system also correctly denied access for an unauthorized face, an outsider's voice and a mismatch between two registered members. These results demonstrate that the pipeline functions as intended, although the thresholds and performance estimates remain experimental because they were derived from a small dataset. Further testing with more participants, unauthorized identities and varied recording conditions would be necessary before production deployment. The current results demonstrate a functional academic prototype. Additional participants, recording conditions and independent test data would be required before the system could be considered suitable for practical use.

12. References

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